

HOTEL BOOKING CANCELLATION ANALYSIS

A Data Analytics Project Report

Submitted in partial fulfillment of the requirements for the subject **DATA ANALYTICS (DA)**

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ABSTRACT

Hotel booking cancellations have become an important concern for the hospitality industry. When customers cancel their reservations, hotels may experience revenue losses, inefficient room management, and difficulties in planning their operations. The project titled “Hotel Booking Cancellation Analysis” aims to study hotel reservation data and identify patterns associated with booking cancellations. The analysis will focus on factors such as hotel type, lead time, customer type, deposit type, market segment, and previous booking information, depending on the availability of data. Data analytics techniques will be used to organize, examine, and interpret the dataset. Graphs and charts will help present the findings in a clear and understandable manner. The analysis may help identify customer booking patterns and factors associated with a higher likelihood of cancellation. The ultimate objective is to derive useful insights that can support hotels in improving reservation management, reducing avoidable cancellations, and making better business decisions. The findings and recommendations will be based on the selected dataset and the results obtained from the analysis.

Keywords: Data Analytics, Hotel Reservations, Booking Cancellation, Data Visualization, Customer Behavior, Business Intelligence.

1. INTRODUCTION

The hotel industry plays an important role in tourism and hospitality. Hotels receive reservations from customers through online travel agencies, hotel websites, travel agents, and other booking channels. However, not every reservation results in an actual stay. Some customers cancel their bookings before arrival, which creates challenges for hotel management.

Hotel Booking Cancellation Analysis is a data analytics project that focuses on examining hotel reservation data to understand booking cancellation patterns. Data analytics involves collecting, organizing, processing, and interpreting data to discover useful information and support decision-making.

This project aims to analyze different factors that may be associated with booking cancellations, including hotel type, lead time, customer type, deposit type, market segment, and previous booking history, where these variables are available in the dataset.

The project uses tables, statistical summaries, and graphical representations to make the data easier to understand. Visualizations such as bar charts, pie charts, and line graphs can help compare cancellation rates and identify patterns across different booking categories.

The insights obtained from this analysis may help hotel managers improve reservation planning, understand customer booking behavior, and develop strategies to reduce avoidable cancellations. The project also demonstrates how data analytics can be applied to a real-world business problem.

2. PROBLEM STATEMENT

Hotel booking cancellations can affect room availability, revenue planning, and the operational efficiency of hotels. When reservations are cancelled, hotels may have limited time to offer the rooms to other customers.

The main problem addressed by this project is the lack of clear insights into booking cancellation patterns and the factors associated with them. By analyzing historical reservation data, the project seeks to determine how cancellation rates vary across hotel types, booking lead times, customer categories, and other available variables.

The analysis will use data summaries and visualizations to identify meaningful patterns and provide practical recommendations for better reservation management.

3. OBJECTIVES OF THE PROJECT

1. To analyze hotel booking data and understand the overall reservation pattern. 2. To calculate the number and percentage of cancelled and non-cancelled bookings. 3. To compare cancellation patterns between different hotel types.
4. To study the relationship between lead time and booking cancellations. 5. To examine cancellation patterns across customer types, deposit types, and market segments, where data is available.
6. To create graphs and charts for effective data visualization.
7. To identify important trends and patterns in historical booking data. 8. To suggest data-driven measures that may help hotels improve reservation management.

4. DATASET DESCRIPTION

The Hotel Booking Demand dataset is a historical dataset containing hotel reservation information for a city hotel and a resort hotel. It includes both cancelled and non-cancelled bookings.

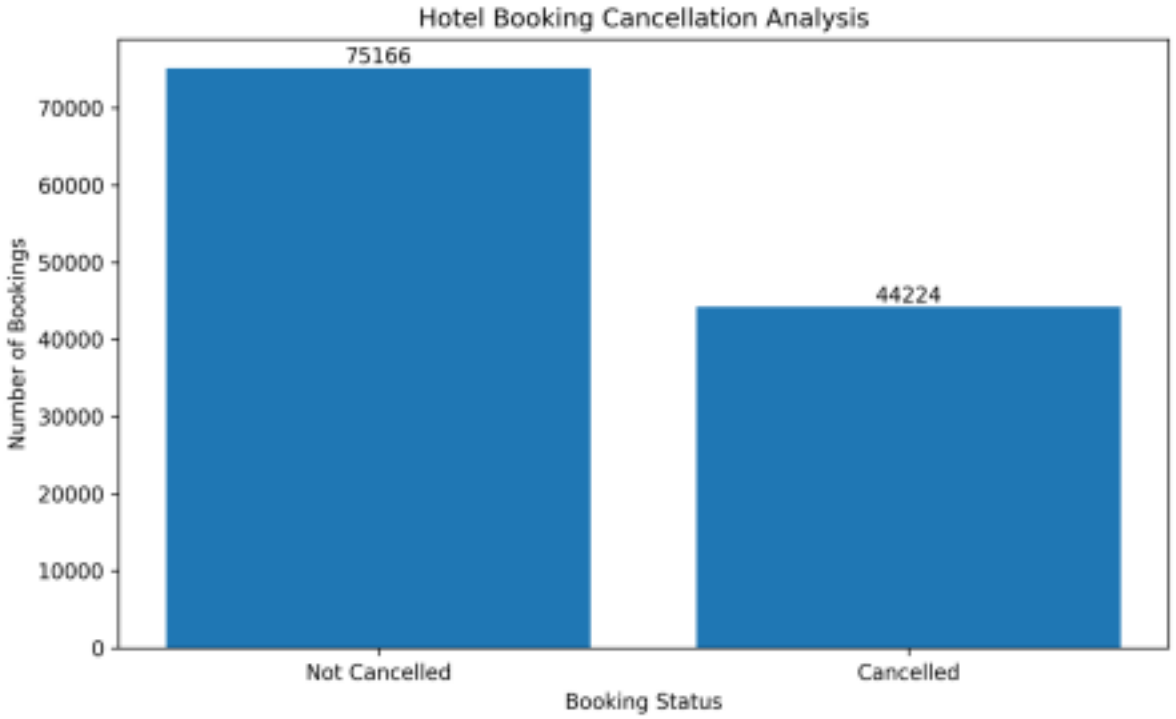
The dataset contains 119,390 booking records and 32 columns. The records cover arrivals scheduled between July 2015 and August 2017. The data includes information about booking dates, length of stay, number of guests, customer type, deposit type, market segment, and reservation status.

The dataset is suitable for exploratory data analysis because it contains several variables that can be compared to understand booking cancellation patterns. The main objective is to identify trends and relationships between booking characteristics and cancellation outcomes.

Dataset source: Jesse Mostipak, Hotel Booking Demand Dataset, Kaggle.

FORMULA-:

$$CancellationRate = \frac{CancelledBookings}{TotalBookings} \times 100$$



4.1 Analysis of Booking Cancellation Status

The first graph represents the number of cancelled and non-cancelled hotel bookings. It provides an overview of the booking status distribution in the selected dataset. By comparing both categories, we can understand how frequently customers cancel their reservations. The cancellation percentage is calculated by dividing the number of cancelled bookings by the total number of bookings and multiplying the result by 100.

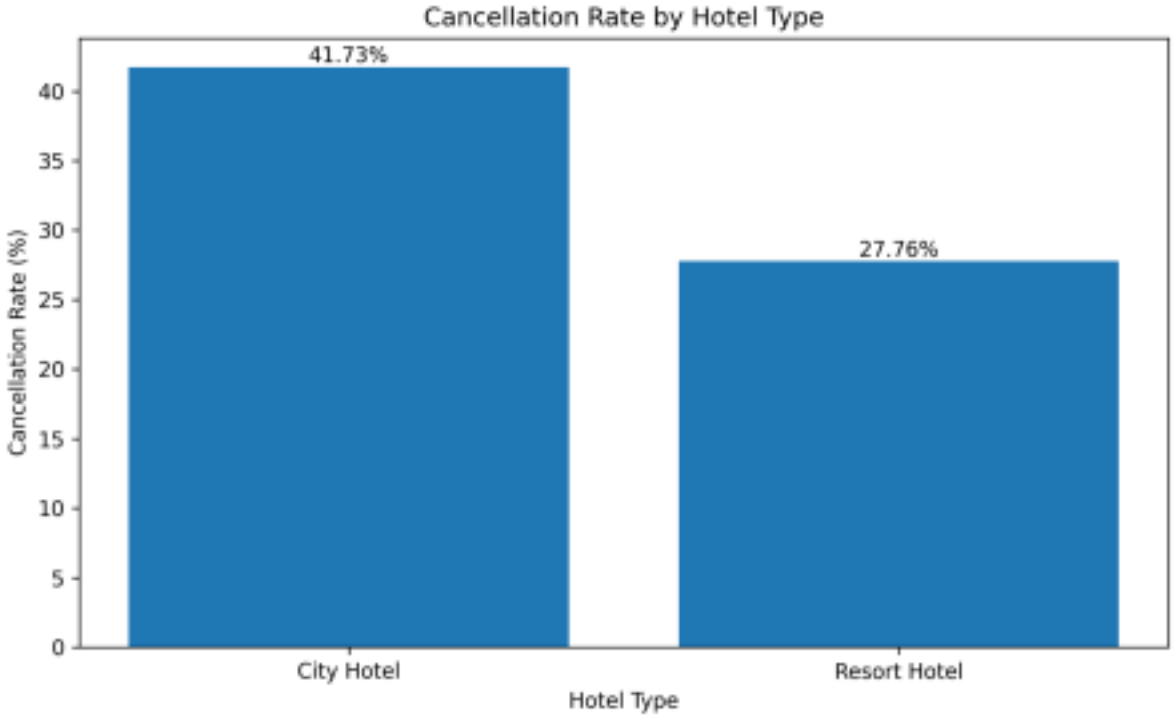
This analysis provides a foundation for studying the factors associated with hotel booking cancellations in the following sections.



4.2 Hotel Type-wise Cancellation Analysis

This analysis compares booking cancellations between City Hotels and Resort Hotels. The grouped bar chart displays cancelled and non-cancelled reservations for each hotel type.

The comparison helps identify which hotel category has a greater number of cancellations. However, the total number of bookings may differ between the two hotel types, so cancellation percentages should also be calculated for a fair comparison.

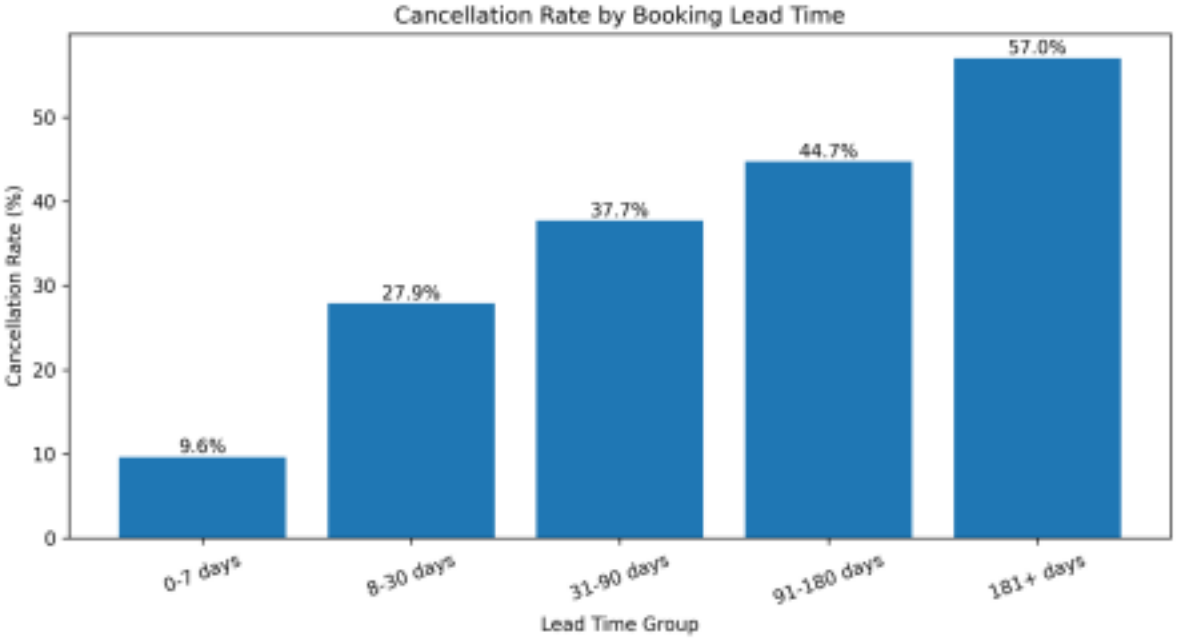


4.3 Comparative Cancellation Rate Analysis

The cancellation rate is calculated separately for City Hotels and Resort Hotels by dividing the number of cancelled bookings by the total number of bookings in each category.

This percentage-based comparison provides a fairer assessment than comparing cancellation counts alone because the two hotel categories may have different numbers of reservations.

The results help identify which hotel type has a higher cancellation rate in the selected dataset. These findings can support further investigation into customer booking patterns and reservation management.



4.4 Lead Time Analysis

Lead time refers to the number of days between the date of booking and the scheduled arrival date. This analysis groups reservations into different lead-time categories and

calculates the cancellation rate for each group.

The bar chart helps compare cancellation rates among customers who booked shortly before arrival and those who booked well in advance. The results can help identify whether booking lead time is associated with cancellation behavior.

The findings will be interpreted using the actual values obtained from the dataset. A relationship between lead time and cancellation does not necessarily mean that lead time directly causes a cancellation.

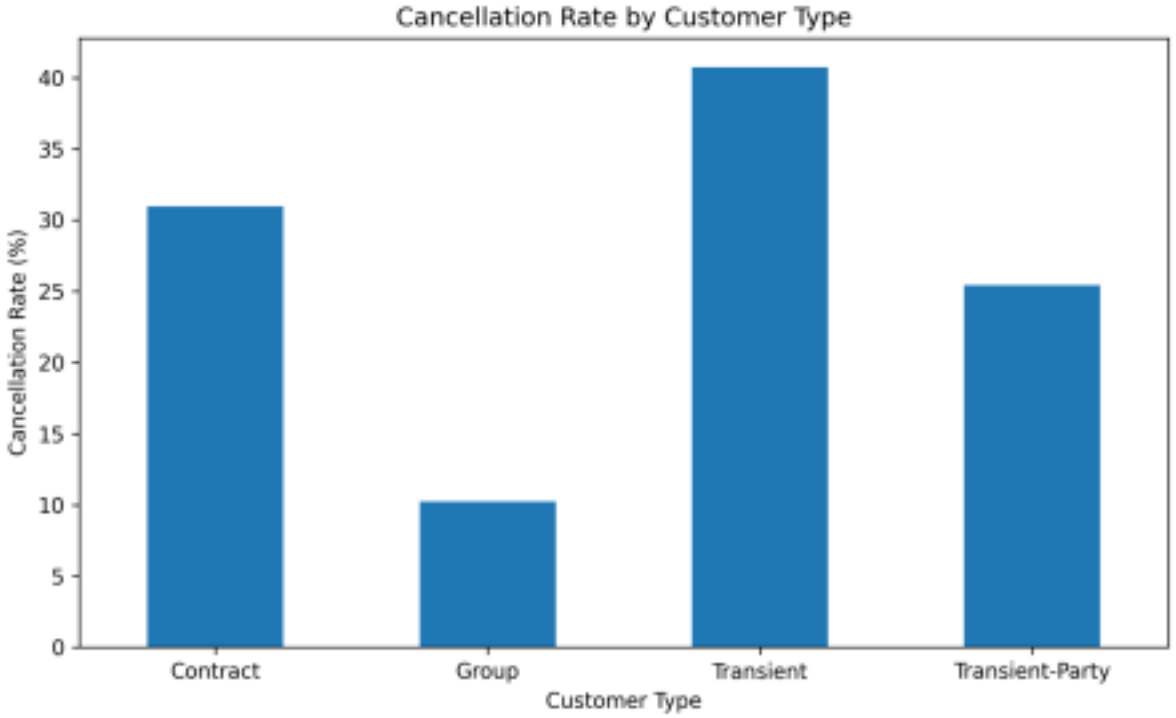


4.5 Monthly Cancellation Analysis

This analysis examines hotel booking cancellation rates across different arrival months. The line graph presents the monthly cancellation rate calculated from the number of cancelled bookings divided by total bookings for each month.

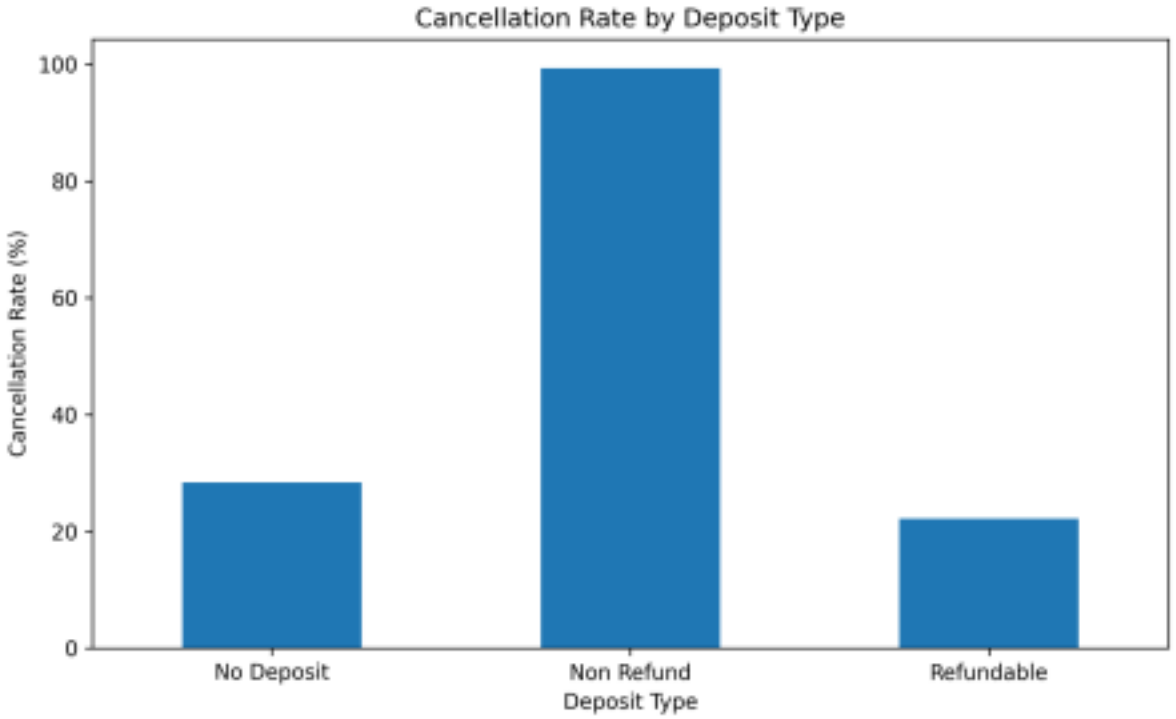
The visualization helps identify seasonal variations and months with relatively higher or lower cancellation rates. Such patterns may help hotel managers plan room availability and reservation strategies.

The results represent the months included in the selected dataset and should not automatically be treated as a forecast for future years.



4.6 Customer Type Analysis

This analysis studies hotel booking cancellations across different customer types. The bar graph compares cancellation rates for different categories of customers. It helps identify which customer groups have relatively higher cancellation rates. Hotel managers can use this information to improve customer policies and reservation management. The findings are based on the available dataset.



4.7 Deposit Type Analysis

This analysis examines the relationship between deposit type and booking cancellation. The graph compares cancellation rates for different deposit policies, such as no deposit, non-refundable, and refundable bookings. The results help hotel managers understand how deposit policies are associated with cancellations. These findings show an association and do not necessarily prove that deposit type causes cancellations.

5. Results and Findings

The analysis of the Hotel Booking Demand dataset provides useful insights into hotel booking cancellations.

1. Overall Cancellation: The dataset analysis shows the proportion of cancelled and non-cancelled bookings.
2. Hotel Type: Cancellation rates are compared between City Hotels and Resort Hotels to identify differences in booking behaviour.
3. Lead Time: Bookings are grouped according to the number of days between reservation and arrival. This helps examine how lead time is associated with cancellations.
4. Monthly Trends: Monthly cancellation rates are analysed to understand possible seasonal variations.
5. Customer Type: Different customer categories are compared to identify variations in cancellation behaviour.
6. Deposit Type: Cancellation rates are compared across deposit policies to understand their association with booking cancellations.

Overall, the visualizations make it easier to understand booking patterns and identify areas where hotel management may improve reservation planning.

6. Recommendations

Based on the analysis, the following recommendations can help hotel managers:

- Improve Reservation Policies: Review booking conditions for customer groups with higher cancellation rates.
- Manage Advance Bookings: Monitor bookings made far in advance and consider appropriate reminder strategies.
- Use Seasonal Planning: Study monthly booking trends to plan room availability and staffing.
- Review Deposit Policies: Evaluate deposit rules carefully while considering customer experience and cancellation patterns.
- Monitor Booking Data: Regularly analyse booking records to identify changing trends and support better decisions.
- Communicate with Customers: Send booking confirmations and reminders to help customers manage their reservations.

7. Conclusion

The project, “Hotel Booking Cancellation Analysis,” demonstrates how data analytics can be used to study hotel booking behaviour. Using Python, Pandas, Matplotlib, and the Hotel Booking Demand dataset, different aspects of booking cancellations were analysed through tables and graphs.

The project examined cancellation rates by hotel type, lead time, arrival month, customer type, and deposit type. These analyses help present booking patterns in a clear and

understandable form.

This project improved my understanding of data cleaning, grouping, percentage calculation, and data visualization. In conclusion, data analytics can support better hotel reservation planning and informed decision-making. Future work may include predictive models to estimate the likelihood of a booking being cancelled.

References

1. Jesse Mostipak, Hotel Booking Demand Dataset, Kaggle. https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand
2. Pandas Documentation. https://pandas.pydata.org/docs / 3. Matplotlib Documentation. https://matplotlib.org/stable/
4. Google Colab. https://colab.research.google.com/